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Polypropylene Insulation Failure Model Modification Based on Space Charge and Trap Parameters

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ABSTRACT

The performance of polypropylene film determines the reliability of capacitors. This paper studies the change of insulation characteristics of the insulation dielectric polypropylene of high-voltage capacitors during the long-term operation process. Moreover, the correction of the insulation failure model of polypropylene media is implemented. Some conclusions are drawn from a series of experiments. The breakdown strength of the insulation dielectrics decreases as the withstand time rises, with a maximum decrease of 37.91% compared to the polypropylene sample without withstand voltage. As the electric field increases from -20 to -100 kV/mm, the total amount of space charge, the trap energy level and the trap density inside the polypropylene increase by 524.15%, 18.99% and 773.36%, respectively. In the capacitor simulation model, the electric field in the capacitor increases by 19.62% after presetting space charge relative to the nonspace charge. The experimental data are substituted into the inverse power model, and the trap parameters are used to modify the model. The results show that the ageing state of the polypropylene has a negative exponential relationship with the trap parameters. This paper provides an important reference for the study of the failure mechanism of polypropylene in high-voltage capacitors and the assessment of polypropylene ageing state.

1 | Introduction

In recent years, with the continuous progress of economic development and modern industry, the safe and stable operation requirements of high-voltage capacitors have been increasing. The stability of capacitors largely depends on the performance of polypropylene (PP) polymer films [1, 2]. Space charge injection and electric field distortion are inevitable in PP during long-term operation [3]. As a result, PP gradually ages, and even the insulation becomes ineffective. Therefore, it is urgent to study the insulation characteristics and failure mechanisms of electrical equipment for economic and safety reasons [4].

Many scholars have carried out a series of studies on the failure mechanism of polymer films. Abderrazak et al. studied the

criticality of different defects in polypropylene random (PPR) copolymer. The remaining life of the defective pipes was predicted [5]. Cui et al. described the life probability distribution of polypropylene fibre parameters through the two-parameter Weibull distribution. The shape parameters and characteristic life parameters were also obtained [6]. Hur et al. tested different types of PP and proposed a modified strain rate model. The model satisfies the physical parameter dependence and predicts fatigue life data with some accuracy [7]. However, the limitations of the insulation dielectric failure model constructed by concentrating on the external changes of the dielectric are strong. It is difficult to explore deeper failure mechanisms. Therefore, to deeply understand the insulation failure mechanism, the research based on the change of internal microscopic properties of polymers has become a hot topic in the industry.

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Many scholars have carried out a series of research studies on charge accumulation and trap parameter variation of insulation materials. Yan et al. investigated the effect of trap energy level density on the polymer space charge distribution, and it was found that the introduction of traps contributes to carrier transport and thus can reduce the breakdown strength [8]. Dang et al. investigated the effect of space charge accumulation on the deterioration of high-temperature properties of polymeric insulating materials. It was concluded that suppression of space charge is beneficial to maintain dielectric insulation properties [9]. Chen et al. believed the space charge effect and insulation ageing interact with each other. Space charge will accelerate insulation ageing, and more space charge will be generated during insulation ageing [10]. It can be seen that the variation of space charge and trap parameters plays an important role in the failure process of insulation dielectrics. Therefore, the insulation failure can be studied by using space charge and trap parameter variations within the insulating dielectric.

The operation environment of high-voltage capacitors is complex. Therefore, the operation of PP films will be affected by many factors. At present, many ageing state models are constructed only for breakdown performance or space charge trap characteristics effects. Only single factors are taken into account in these models, so their applicability is weak. Therefore, PP films are investigated through various aspects such as ageing state models, trap parameters and charge characteristics in order to adapt to various operating environment changes. This experiment starts with the breakdown characteristic of PP film. The comprehensive consideration of the influence of external field strength on the internal space charge and trap characteristics of PP samples is undertaken, and the classical inverse power model is modified.

In this paper, a series of studies are carried out on the insulation properties of PP films for capacitors. Firstly, the breakdown performance of PP samples is tested by the step-up method, and the inverse power model is established. Secondly, the space charge parameters and trap characteristics of PP samples under different electric ageing fields are studied. The three-dimensional relationships of field strength, charge and trap parameters are established. Finally, space charge and trap parameters are introduced into the classical inverse power model. The inverse power model of PP is modified based on trap characteristics.

2 | Sample Preparation and Experimental Method

2.1 | Sample Preparation

PP film samples were prepared using a flat vulcanising machine. Firstly, the PP particles were placed in a 180°C flat vulcanising machine for pressurised melting treatment (20 MPa, 15 min). Subsequently, the sample was cooled to indoor temperature, resulting in the formation of a 100- μ m PP film. Finally, the sample underwent drying in an oven at 60°C for 30 min.

2.2 | Experimental Methods

According to IEC 60243-1:2013, the breakdown test of PP film samples was carried out using a voltage breakdown tester. The

voltage boost rate was set to 500 V/s, and the temperature was indoor temperature (approximately 25°C). In this experiment, the sample and electrode were completely immersed in silicone oil to prevent surface discharge.

Constant voltage and step-voltage were commonly used in electrical ageing acceleration experiments. Electrical ageing acceleration was achieved by configuring the initial voltage (U_0), step size of voltage increment (ΔU) and duration of withstand voltage (Δt) for each stage. The step-voltage schematic is shown in Figure 1, where U represents the applied voltage, t denotes the voltage application time until the sample broke down, U_n is the voltage at which the dielectric broke down, Δt_i refers to the voltage application time at each level and Δt_m is the time required for the last voltage increase to break down. The test procedure was as follows.

Firstly, a short-time breakdown test was performed on the sample. The breakdown voltage U_b of the sample was obtained. Secondly, the initial voltage U_0 and voltage application time Δt for each stage were determined through extensive preliminary experiment. Thirdly, according to the different Δt , the samples were divided into six groups. Each group contained at least five samples for repeated experiments. Finally, the initial voltage was determined to be 10.5 kV. The step size of voltage increment was 1.75 kV. The voltage application times Δt of each stage in the six groups were 10, 60, 100, 600, 1200 and 1800 s.

Trap parameters of PP specimens were characterised by thermally stimulated depolarising current (TSDC). First, the electric field strength was set to -20, -40, -60, -80 and -100 kV/mm, respectively. The heating rate was 10 K/min. The samples were polarised with different electric fields. Then, the relationship of depolarising current with temperature was tested.

3 | Experimental Results and Analysis

3.1 | Breakdown Characteristics Analysis

The Weibull distribution of two parameters of PP under the action of different voltage application times (Δt) is shown in

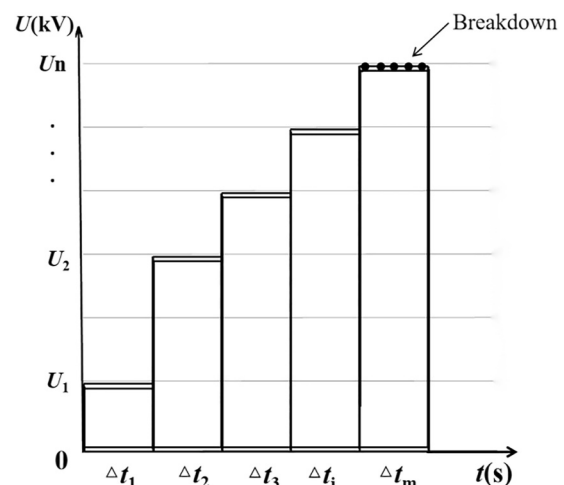


FIGURE 1 | Step-voltage electric ageing.

Figure 2. The scale parameters (α), shape parameters (β) and breakdown strength of samples with different Δt are shown in Table 1.

It can be seen from Figure 2 that the breakdown strength of PP dielectric shows a decreasing trend with the increase in Δt . Furthermore, the declining rate is slowing down. With Δt increasing from 0 to 1800 s, the breakdown strength decreased from 320.00 to 198.66 kV/mm, dropping by 37.91%. It is analysed that the applied field strength is applied to the sample for a longer time as Δt increases. When a DC voltage is applied to the electrode, carriers are injected into the PP by overcoming the potential barrier from the metal electrode. On the one hand, the detrapped free charge collides with the atomic or molecular structure in the form of high-energy electrons, causing damage is caused inside the PP. On the other hand, the free charge inside the PP increases and accumulates near the electrode to form a space charge layer. Local electric field distortions exacerbate breakdown phenomena when the generated local field is strong enough [11, 12]. It is generally believed that the external environment can cause irreversible damage to insulating materials. The internal damage of PP material increases with the duration of the applied voltage. As a result, breakdown is more likely to occur as the applied voltage increases, leading to a significant reduction in breakdown strength [13].

3.2 | Extraction and Analysis of Trap Parameters

In order to study the effect of electric field on the charge accumulation characteristics and trap distribution, the PP samples are polarised by applying different voltages, and the charge released under thermal stimulation is characterised by TSDC.

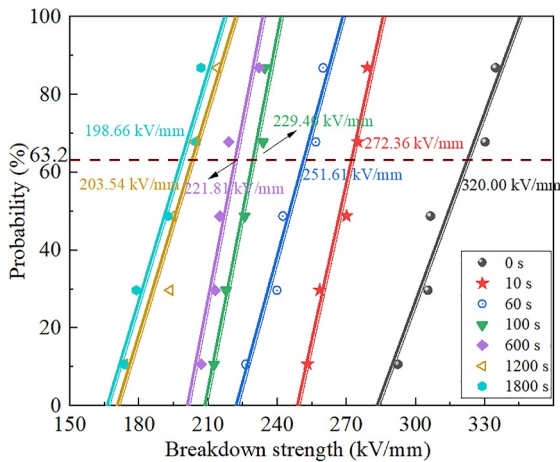


FIGURE 2 | Weibull distribution of samples with different voltage application times.

TABLE 1 | The Weibull distribution parameters of the breakdown of PP with different Δt .

Δt (s)	0	10	60	100	600	1200	1800
α (kV/mm)	320	272.36	251.61	229.49	221.81	203.54	198.66
β	5.076	7.129	5.250	6.792	6.759	3.727	3.730

The TSDC curve of PP varying with temperature is displayed in Figure 3. Its polarisation electric field ranges from -20 to -100 kV/mm, and its interval is 20 kV/mm.

As can be seen from Figure 3, TSDC current amplitude increases with rising polarisation voltage. The polarisation voltage increases from -20 to -100 kV/mm, and the corresponding current amplitude rises from -20 to -108 pA, which increases by 4.40 times. The analysis shows that the polarisation process injects charge into PP. The injection effect is more obvious when the polarisation field intensity is higher. Therefore, the change in depolarisation current is consistent with the polarisation field strength. In addition, the currents corresponding to different polarisation electric fields show a consistent trend with the increase in temperature, reaching a peak at approximately 336 K. It is found that the phenomenon is related to the dynamic process of charge trapping and detrapping [14]. Charge is difficult to escape from the traps at low temperatures, but it easily obtains energy and detraps to form free charge under the action of thermal stimulation [15, 16]. Therefore, the depolarisation current increases.

Through the theoretical calculation of the curve in Figure 3, the charge quantity and trap parameters of PP samples under different polarisation electric fields are extracted. The formulas are shown in Equations (1–5), and their calculation results are shown in Table 2.

According to Debye theory [17], the calculation formula of depolarisation current is shown in Equation (1):

$$I(T) = A \exp \left[-\frac{E_T}{kT} - \frac{1}{\tau_0 \gamma} \int_{T_0}^{T_c} \exp \left(-\frac{E_T}{kT} \right) dT \right], \quad (1)$$

where $I(T)$ is the depolarisation current curve, A; A is a constant; τ_0 is the time constant, s; γ is the heating rate, K/min; k is

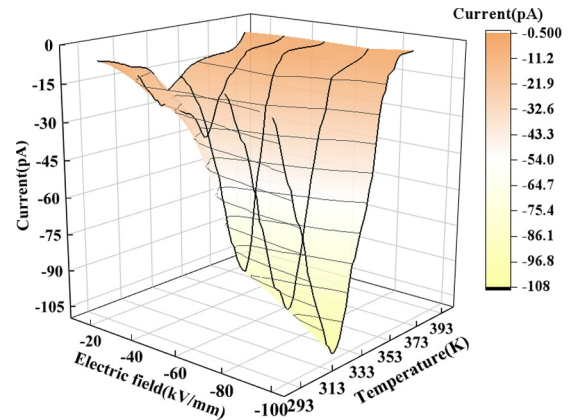


FIGURE 3 | TSDC curves of PP under different polarisation electric fields.

TABLE 2 | The charge quantity and trap parameters of PP under different polarised electric fields.

Applied electric field (kV/mm)	Trapped charge (C)	Trap level (eV)	Trap density (m^{-3})
-20	4.39×10^{-9}	0.79	6.87×10^{17}
-40	7.74×10^{-9}	0.80	1.19×10^{18}
-60	1.55×10^{-8}	0.83	2.36×10^{18}
-80	2.16×10^{-8}	0.86	3.92×10^{18}
-100	2.74×10^{-8}	0.94	6.00×10^{18}

the Boltzmann constant, 8.628×10^{-5} eV/K; T_0 and T_e are the starting and ending temperatures of the current relaxation peaks, respectively, K; and E_T is the trap energy level, eV.

The single-level trap parameters of PP can be determined by the TSDC curve. The charge quantity (Q_{TSDC}) and trap energy level (E_T) can be obtained by Equations (2) and (3), respectively:

$$Q_{\text{TSDC}} = \frac{60}{\gamma} \int_{T_0}^{T_e} I(T) dT, \quad (2)$$

$$E_T = \frac{2.47 k_B T_m^2}{\Delta T}, \quad (3)$$

where $I(T)$ is the TSDC current, A; T_m is the temperature corresponding to the peak current, K; and ΔT is the difference between the temperature and the half-peak of the current.

The depolarisation current density and trap density (N_t) can be calculated by Equations (4) and (5):

$$j(T) = \frac{I(T)}{S}, \quad (4)$$

$$N_t = \frac{1}{\gamma e d} \int_{T_0}^{T_e} j(T) dT, \quad (5)$$

where $j(T)$ is the current density, A/m^2 ; S is the electrode area, m^2 ; d is the sample thickness, m; and e is the elementary charge, 1.60×10^{-19} C.

As shown in Table 2, the charge quantity inside the sample increases with the rise of the applied electric field. The charge quantity increases from 4.39×10^{-9} C at -20 kV/mm to 2.74×10^{-8} C at -100 kV/mm. The analysis shows that it is easier for charge in a high field to cross the interface barrier and inject into PP material because it obtains more energy. This results in higher charge accumulation and charge density.

It is clear from Figure 4 that the variation of trap density with electric field in Table 2 is depicted. It can be seen that the trap density increases exponentially with the increase of electric field.

This can be observed from Figure 4 that the trap energy level of PP increases from 0.79 to 0.94 eV with the increase of polarisation electric field. Trap density increases from $6.87 \times 10^{17} \text{ m}^{-3}$ to $6.00 \times 10^{18} \text{ m}^{-3}$, showing an increase of 773.36%. The reason for this is that the inner crystalline region of PP coexists with the amorphous region. The folding and breaking of the internal molecular chains will produce various

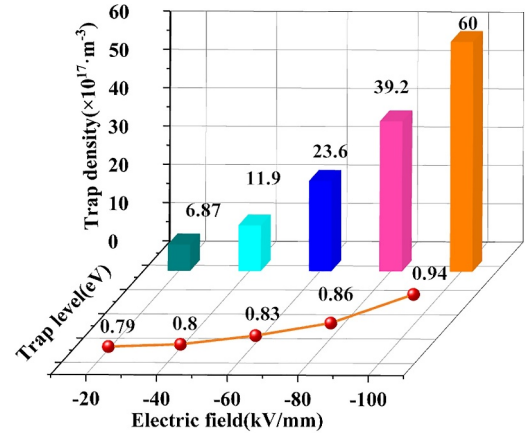


FIGURE 4 | Trap density distribution with electric field variation.

forms of defects. The increase of the external electric field promotes the ionisation of the internal impurities and the movement of the molecular chains [18, 19]. Therefore, the trap density and trap energy level in the amorphous region increase.

3.3 | Charge Characteristic Simulation Analysis

In order to further analyse the variation characteristics of space charge, the structure of a PP film capacitor with both ends coated is constructed using finite element simulation software. The mechanism of space charge in dielectrics with different field strengths is investigated.

In high-voltage capacitors, a metallised film is formed by vacuum evaporation of an extremely thin aluminium layer on the side of the PP film [20]. PP film thickness is generally a few microns. The thickness of the vaporised metal electrode is usually tens of nanometres. The high- and low-voltage electrodes of the capacitor are often plated in a staggered manner by leaving the edge and leading through the gold-sprayed layers on each side of the capacitor.

The effect of the electrostatic field as well as the effect of the space charge generated by the electric field should be considered in the calculation. Based on the law of conservation of charge, a body charge is added in the medium by defining the body charge density (ρ_v), as shown in Equation (6):

$$\begin{cases} E = -\nabla V \\ \nabla \cdot (\epsilon_0 \epsilon_r E) = \rho_v \end{cases} \quad (6)$$

where ϵ_0 is the dielectric constant of vacuum, usually a value of $8.854187817 \times 10^{-12}$ F/m. ϵ_r represents the relative dielectric constant of PP; usually a value of 2.1 is selected.

The thickness of the metallised film is generally much less than the radius of the mandrel. Therefore, the radial direction of the circular arc metallised film element is set to be parallel in a region with a small arc length. In the simulation study, the relative dielectric constant of PP film is 2.2. The volume resistivity is 5.8×10^{17} Ω -cm. The metal layer on the surface is aluminium. The electric field distribution is divided into two regions based on the structural characteristics of the model: one is the boundary region on both sides of the component and the other is the uniform area in the middle of the component.

High-voltage capacitors operate at extremely high field strengths compared to other insulating structures. Therefore, space charge accumulation and partial discharge are likely to occur. The electric field distortion in the capacitor under different electric fields is presented in Figure 5. It can be observed that under the

action of the electric field, the dielectric edges, as well as the areas near the metal electrodes, produce significant distortions.

The electric field distribution in the insulation dielectrics can be significantly distorted by high DC voltages. It also directly affects the ageing and breakdown properties of materials [21]. As can be seen in Figure 5, the electric field distribution is sharply affected by the magnitude of the applied electric field. When the -20 kV/mm electric field is applied, the maximum electric field distortion becomes -51.90 kV/mm. The electric field increases by 159.50%. Space charge is generated within PP during the operation of the applied electric field. Therefore, space charge is added at different electric fields according to the trapped charges, as shown in Table 2. The space charge distribution at the cross-section is presented in Figure 6.

Some electric fields will be superimposed in the presence of space charge. Therefore, the distortion is obvious and the actual electric field is much higher than the applied electric field. At -20 kV/mm, the maximum electric field distortion increases

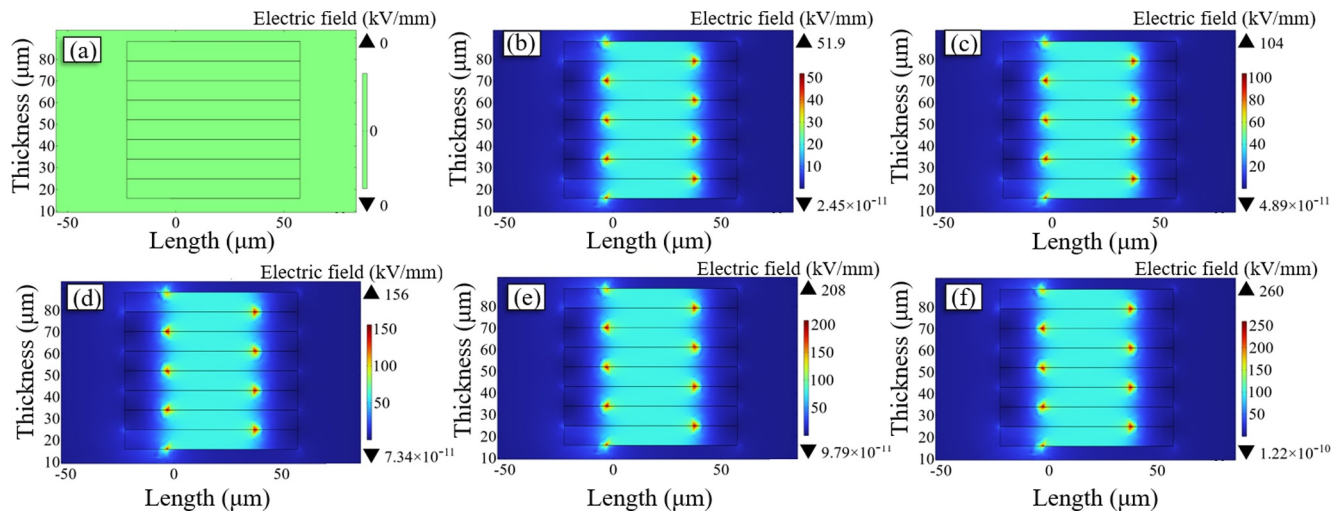


FIGURE 5 | Electric field vector modulus distribution in a capacitor without space charge under different applied field strengths: (a) 0 kV/mm, (b) -20 kV/mm, (c) -40 kV/mm, (d) -60 kV/mm, (e) -80 kV/mm, (f) -100 kV/mm.

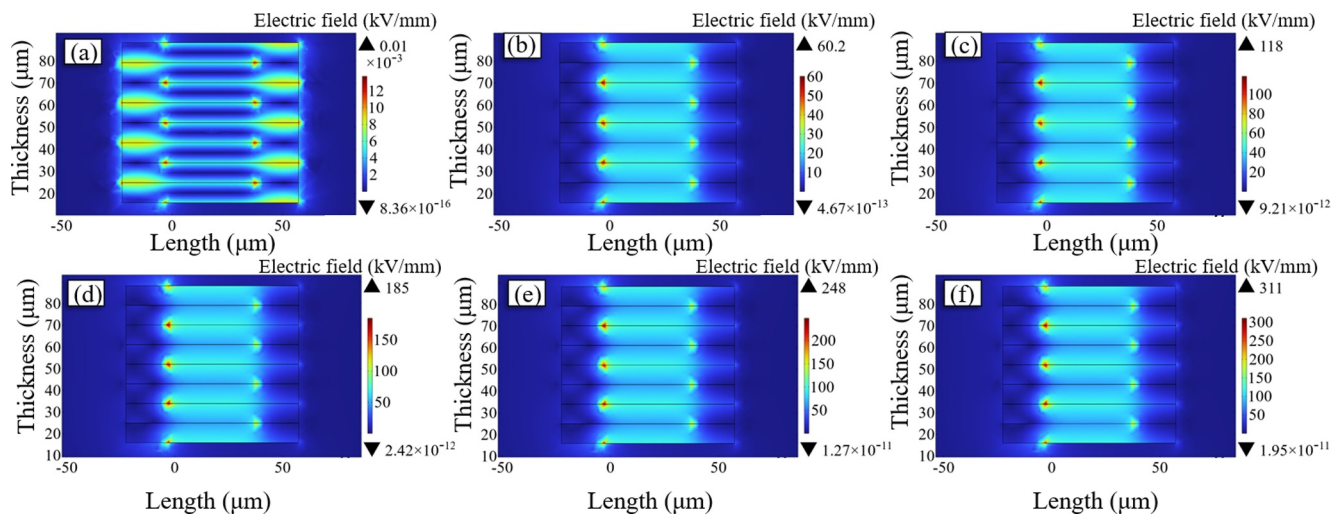


FIGURE 6 | Electric field vector modulus distribution in a capacitor with space charge under different applied field strengths: (a) 0 kV/mm, (b) -20 kV/mm, (c) -40 kV/mm, (d) -60 kV/mm, (e) -80 kV/mm, (f) -100 kV/mm.

from -51.90 to -60.2 kV/mm, increasing by 15.99%. As the field strength becomes -100 kV/mm, the maximum distortion field rises to -311 kV/mm, which increases by 19.62%. It can be observed that the effect of space charge on the distortion of the electric field is significant. Therefore, exploring the change in space charge within a certain field strength is an important parameter to judge the electrical ageing degree of the dielectrics.

4 | Inverse Power Model Modification Based on Trap Parameters

4.1 | Establishment of Inverse Power Model

It is believed that solid insulation is irrecoverable by external action. In other words, dielectric insulation failure has a cumulative effect. The ageing model is based on the classical inverse power model (IPM) with modifications [22]. As shown in Equation (7), it is the inverse power model:

$$U^n t = D. \quad (7)$$

Take logarithms of both sides of Equation (7) to obtain Equation (8):

$$\ln t = -n \ln U + \ln C, \quad (8)$$

where U is the voltage applied to the dielectrics, kV. t is the time taken by the dielectrics to break down under this voltage, s. D , C and n are constants and depend on the dielectrics itself. n is also known as the life index. D is also regarded as the damage value.

The equivalent breakdown voltage and equivalent breakdown time of different Δt are obtained by step-voltage experiments. The slope of the line in the range n is obtained by least square fitting. Decomposing the inverse power model, the voltages U_i at each level and the durations t_i at each level are substituted, $i = 1, 2 \dots m$; m represents the number of times voltage is applied step by step during breakdown, as shown in Equation (9):

$$t_1(U_1)^n + t_2(U_2)^n + t_3(U_3)^n + \dots + t_m(U_m)^n = D, \quad (9)$$

where U_m denotes the voltage at which breakdown occurs, kV. t_m denotes the duration at this voltage, s. It should be noted that the duration of each stage is the same in one experiment.

The samples are divided into six groups with Δt of 10, 60, 100, 600, 1200 and 1800 s. Each group contains at least five samples for repeated experiments. Plugging the data into the calculation, the critical damage D_{ab} can be obtained. The obtained damage matrix D is shown as Equation (10):

$$D = \begin{bmatrix} D_{11} & D_{12} & \dots & D_{1b} \\ D_{21} & D_{22} & \dots & D_{2b} \\ \vdots & \vdots & \vdots & \vdots \\ D_{41} & D_{42} & \dots & D_{4b} \\ \vdots & \vdots & \vdots & \vdots \\ D_{a1} & D_{a2} & \dots & D_{ab} \end{bmatrix}, \quad (10)$$

where D_{ab} represents the damage values obtained by respectively introducing different voltages and times, and $a = 1, 2 \dots 6$ and $b = 1, 2 \dots 5$.

The following assumptions are made in this model. The life index of the sample is a constant value and it is independent of the applied voltage. In the same material, the damage is a constant value from the initial voltage to the final breakdown.

Therefore, the D_{ab} in the damage matrix of the PP electrical ageing test should be equal. However, there are unavoidable differences in specimen preparation and experimental testing. The result is uncertain and the D_{ab} of each experiment will not be exactly equal. In addition, the choice of n value also has a significant influence on D_{ab} . When the given n value is close to the actual n value, the dispersion of D_{ab} will be small [23]. Therefore, the minimum dispersion of D_{ab} is the true value of n . The dispersion of D_{ab} can be represented by the variance $S^2(n)$, which is shown in Equation (11):

$$S^2(n) = \frac{\sum_{a=1}^6 \sum_{b=1}^5 (D_{ab} - \bar{D})^2}{6 \times 5 - 1}, \quad (11)$$

$\bar{D}$ is the average value of D_{ab} .

For this reason, choices can be made in the range of n values to ensure accuracy.

4.2 | Trap Parameter Modification

It can be found in the TSDC experiment that the charge accumulation inside the dielectric increases with the applied electric field. The trap energy level and trap density also change at different electric fields. The trap density shows an exponential change with the increase in electric field [24]. This internal connection can relate the electrical ageing of PP to the trap energy level and trap density of the dielectrics.

Within a certain field strength range, the trap energy level inside PP is affected by the applied voltage. The relationship between trap energy level and applied voltage is given in Equation (12):

$$U \propto E_t^{f_1}, \quad (12)$$

where E_t is the trap energy level of the dielectric, eV; and f_1 is the coefficient. At the same time, the inner trap density also changes during the ageing process of PP.

The relationship between trap density and applied voltage is shown in Equation (13):

$$U \propto N_t^{f_2}, \quad (13)$$

where N_t is the trap density inside the dielectrics, m^{-3} ; and f_2 is the coefficient.

The ageing model of PP is related to the applied voltage. Therefore, by combining Equations (12) and (13), Equation (14) can be obtained:

$$D \propto U^{-n} \propto E_t^{f_1} \propto N_t^{f_2}. \quad (14)$$

In addition, the relationship between electrical ageing and voltage of PP can be obtained by Equation (15):

$$D \propto E_t^{f_1} N_t^{f_2}. \quad (15)$$

It is evident from Equation (15) that the ageing model of PP presents an exponential relationship.

The relationship of trap energy level and trap density with voltage is derived from TSDC data. The trap parameters at different field strengths are calculated by substituting them into Equation (15). The trap energy level and trap density dependence functions of PP at different electric fields can be obtained. The final parameter fitting diagram is shown in Figure 7.

The relationship between the final electric field and the trap parameter is given by Equation (16):

$$E = A_1 \exp(t_1 \cdot E_t + t_2 \cdot N_t') + E_0. \quad (16)$$

In the formula, the value of N_t' is the actual measured trap density divided by 10^{17} . The A_1 , t_1 , t_2 and E_0 are constants.

To sum up, the constant D , life index n and a series of trap parameters in Equation (7) are obtained using the processing methods given in this experiment. The values are substituted into the equation to obtain Equation (17):

$$[A_1 \exp(t_1 \cdot E_t + t_2 \cdot N_t') + E_0]^n d^n t = D. \quad (17)$$

The modified model can be used to estimate the life of PP insulation materials under certain trap conditions.

With the increase in high-voltage capacitor running time, the accumulation of space charge in PP increases. The linear relationship between the variation of trap characteristics and running time becomes more evident. This corresponds to the

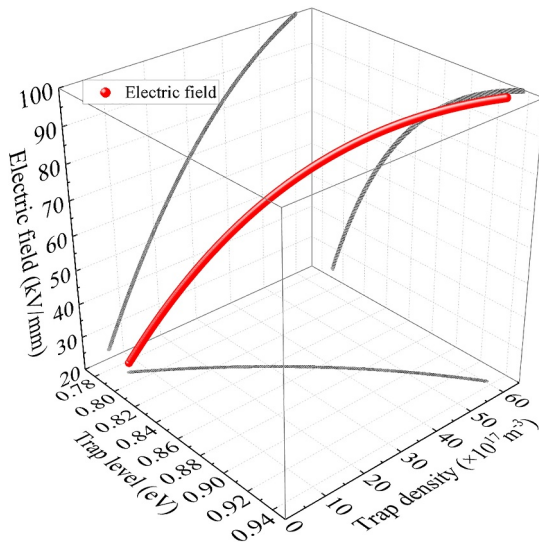


FIGURE 7 | Relationship between electric field and trap parameters.

decrease in breakdown strength during the ageing process. Therefore, it is feasible to evaluate the operating state of capacitors using trap changes in PP dielectrics.

5 | Conclusion

This paper is based on the variation of space charge and trap characteristics in polymer film PP of high-voltage capacitor, and the failure model is modified. The main conclusions reached are as follows:

1. The step-voltage experiment of PP film is carried out, and the voltage application time of each stage is different. The average breakdown strength of each group of samples decreases with the increase in Δt . The maximum decline is approximately 37.91%. The breakdown strength and time are substituted into the inverse power model to provide data support for model correction.
2. The space charge, trap energy levels and densities in the PP increase as the externally applied field strength rises due to the internal structure of the dielectric and the charge injection properties. The trap energy level increases by 18.99%, and the trap density increases by 773.36%. Within a certain range of electric field, the trap density changes exponentially with the increase in electric field.
3. Under different electric fields, there will be different degrees of electric field distortion in high-voltage capacitors. The space charge exacerbates changes in the electric field, and the effect is more severe at low field strength. The maximum distortion electric field is -51.90 kV/mm under the applied -20 kV/mm electric field, which increases by 159.50%. The worst electric field distortion increases 19.62% at the same electric field after presetting the space charge.
4. The failure change of PP is a rather complex process. Because of the effect of the electric field, the state change of PP has a negative exponential relationship with the trap energy level and trap density. The failure model of this PP material centres on the changes in space charge and trap characteristics. The data dispersion of the model is small, and its reliability is high. It is applicable to the study and analysis of the performance and failure mechanism of solid insulating dielectrics.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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